CHAPTER EXERCISE: Answers for this chapter start on page 309.

A calculator should NOT be used on the following questions.

1

The water level h, in feet, in a large aquarium can be modeled by $h = 100 - 3d$, where d is the number of days that have passed since the aquarium was last refilled. Based on the model, how does the water level change each day?

A) Decreases by 3 feet
B) Increases by 3 feet
C) Decreases by 100 feet
D) Increases by 100 feet

2

The number of loaves of bread b remaining in a bakery each day can be estimated by the equation $b = 200 - 18h$, where h is the number of hours that have passed since the store's opening. What is the meaning of the value 18 in this equation?

A) The bakery sells all its loaves of bread in 18 hours.
B) The bakery sells 18 loaves of bread each hour.
C) The bakery sells a total of 18 loaves of bread each day.
D) There are 18 loaves of bread left in the bakery at the end of each day.

3

A membership website offers video tutorials on programming. The number of members, m, subscribed to the site can be estimated by the equation $m = 500 + 200n$, where n is the number of videos available on the site. Based on the equation, which of the following statements is true?

A) For every one additional video, the site gains 500 new members.
B) The site initially made 200 videos available to members.
C) The site was able to get 500 members without any available videos.
D) The site gains 500 new members for every 200 additional videos available on the site.

4

$$s = 10 - 2h$$

A recipe suggests sweetening honey tea with sugar. The equation above can be used to determine the amount of sugar s, in teaspoons, that should be added to a tea beverage with h teaspoons of honey. What is the meaning of the 2 in the equation?

A) For every teaspoon of honey in the beverage, two more teaspoons of sugar should be added.
B) For every teaspoon of honey in the beverage, two fewer teaspoons of sugar should be added.
C) For every two teaspoons of honey in the beverage, one more teaspoon of sugar should be added.
D) For every two teaspoons of honey in the beverage, one fewer teaspoon of sugar should be added.

5

The monthly salary of a salesperson at a used car dealership is determined by the expression $1,000 + 2,000xc$, where x is the salesperson's commission rate and c is the number of cars sold by the salesperson. Which of the following statements is the best interpretation of the number 2,000 in the context of this problem?

A) The average price of a used car at the dealership

B) The base monthly salary of a salesperson at the dealership

C) The average monthly commission earned by each salesperson at the dealership

D) The average number of cars sold by the dealership each month

6

$$p = 2,000s + 15,000$$

A state government uses the equation above to estimate the average population p for a town with s schools. Which of the following best describes the meaning of the number 2,000 in the equation?

A) The average number of students at each school in a town

B) The average number of schools in each town

C) The estimated increase in a town's population for each additional school

D) The estimated population of a town without any schools

7

$$h = 100 - 4t$$

The equation above can be used to model the number of hours h until a gallon of milk held at a temperature of t, in degrees Celsius, goes sour. Based on the model, which of the following is the best interpretation of the number 4 in the equation?

A) An increase of $1°C$ will make a gallon of milk go sour 4 hours faster.

B) An increase of $1°C$ will make 4 gallons of milk go sour 1 hour faster.

C) An increase of $4°C$ will make a gallon of milk go sour 1 hour faster.

D) An increase of $4°C$ will make a gallon of milk go sour 4 hours faster.

8

An antique lamp was sold at an auction. The price p of the lamp, in dollars, during the auction can be modeled by the equation $p = 900 - 10t$, where t is the number of seconds left in the auction. According to the model, what is the meaning of the 900 in the equation?

A) The starting auction price of the lamp

B) The final auction price of the lamp

C) The increase in the price of the lamp per second

D) The time it took to auction off the lamp, in seconds

9

$$y = 1.30x - 1.50$$

A bank teller uses the equation above to exchange U.S. dollars into euros, where y is the euro amount and x is the U.S. dollar amount. Which of the following is the best interpretation of the 1.50 in the equation?

A) The bank charges 1.50 euros to do the currency exchange.

B) The bank charges 1.50 U.S. dollars to do the currency exchange.

C) One U.S. dollar is worth 1.50 euros.

D) One euro is worth 1.50 U.S. dollars.

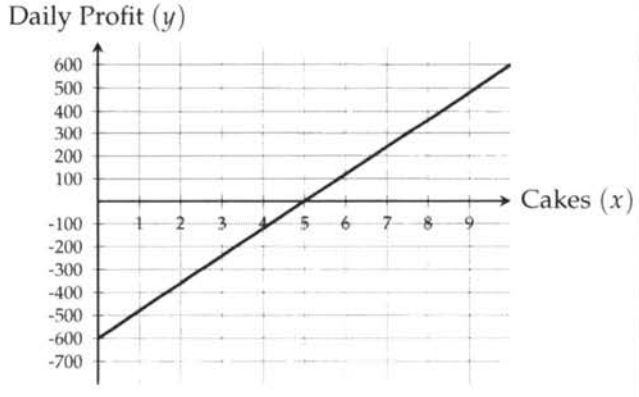
A calculator is allowed on the following questions.

10

$$t = \frac{2x + 9}{5}$$

The equation above models the time t, in seconds, it takes to load a web page with x images. Based on the model, by how many seconds does each image increase the load time of a web page?

Questions 11-13 refer to the following information.



The relationship between the daily profit y, in dollars, of a bakery and the number of cakes sold by the bakery is graphed in the xy-plane above.

11

What does the slope of the line represent?

A) The price of each cake
B) The profit generated from each cake sold
C) The daily profit generated from all the cakes that were sold
D) The number of cakes that need to be sold to make a daily profit of 100 dollars

12

Which of the following is the best interpretation of the y-intercept in the context of this problem?

A) The price of each cake
B) The cost of making each cake
C) The daily costs of running a bakery
D) The daily cost of making the cakes that weren't able to be sold

13

What does it mean that $(5, 0)$ is a solution to the equation of the line?

A) The bakery needs to sell 5 cakes per day to cover its daily expenses.
B) Each cake must be sold for at least 5 dollars to cover the cost of making it.
C) It costs 5 dollars to make each cake.
D) Each day, the bakery gives the first 5 cakes away for free.

14

$$T = 56 + 5h$$

To warm up his room, Patrick turns on the heater. The temperature T of his room, in degrees Fahrenheit, can be modeled by the equation above, where h is the number of hours since the heater started running. Based on the model, what is the temperature increase, in degrees Fahrenheit, for every 30 minutes the heater is turned on?

15

$$2y - x = 14$$

Alice owns a pet frog but would like to add turtles to the same tank. The local veterinarian uses the equation above to determine the total amount of water y, in gallons, that should be held in the tank for x turtles to thrive alongside Alice's frog. Based on the equation, which of the following must be true?

I. One additional gallon of water can support two more turtles.

II. One additional turtle requires two more gallons of water.

III. One more turtle requires an additional half a gallon of water.

A) II only

B) III only

C) I and II only

D) I and III only

16

$$C = 1.5 + 2.5x$$

A local post office uses the equation above to determine the cost C, in dollars, of mailing a shipment weighing x pounds. An increase of 10 dollars in the mailing cost is equivalent to an increase of how many pounds in the weight of the shipment?

A) 2

B) 2.5

C) 4

D) 5

Chapter 15: Interpreting Linear Models

CHAPTER EXERCISE:

1. [A] The slope is -3, which means the water level decreases by 3 feet each day.

2. [B] The value 18 refers to the slope of -18, which means the number of loaves remaining decreases by 18 each hour. This implies that the bakery sells 18 loaves each hour.

3. [C] The y-intercept of 500 means that when $n = 0$ (when there were no videos on the site), there were 500 members.

4. [B] The number 2 refers to the slope of -2, which means two fewer teaspoons of sugar should be added for every teaspoon of honey already in the beverage. Don't be fooled by answers (C) and (D), which "reverse" the x and the y (h and s, in this case). The slope is always the change in y for each unit increase in x, not the other way around.

5. [A] The salesperson earns a commission, but on what? The amount of money he or she brings in. To get that, we must multiply the number of cars sold by the average price of each car. Since x and c already represent the commission rate and the number of cars sold, respectively, the number 2,000 must represent the average price of each car.

6. [C] The number 2,000 refers to the slope, which means a town's estimated population increases by 2,000 for each additional school in the town.

7. [A] The number 4 refers to the slope of -4, which means an increase of 1° C decreases the number of hours until a gallon of milk goes sour by 4. In other words, the milk goes sour 4 hours faster.

8. [B] When $t = 0$, there is no time left in the auction. The auction has finished. Therefore, the 900 is the final auction price of the lamp.

9. [A] Because it's the slope, the 1.30 can be thought of as the exchange rate, converting U.S. dollars into euros. But after the conversion, 1.50 is subtracted away, which means you get 1.50 euros less than you should have. Therefore, the best interpretation of the 1.50 y-intercept is a 1.50 euro fee the bank charges to do the conversion.

10. [.4] To see the answer more clearly, we can put the equation into $y = mx + b$ form: $t = \frac{2}{5}x + \frac{9}{5}$. The slope is $\frac{2}{5}$, or 0.4, which means the load time increases by 0.4 seconds for each image on the web page.

11. [B] The slope is the change in y (daily profit) for each unit change in x (cakes sold).

12. [C] Notice that the y-intercept is negative. It is the bakery's profit when no cakes are sold. Therefore, anything that varies with the number of cakes sold is incorrect. For example, answer (D) is wrong because the cost of the cakes that didn't sell depends on how many the bakery did sell. It's not a fixed number like the y-intercept is. The best interpretation of the y-intercept is the cost of running the bakery (rent, labor, machinery, etc.), which is likely a fixed number.

13. [A] The solution $(5, 0)$ means that the bakery's daily profit is zero when 5 cakes are sold. Therefore, selling five cakes is enough to break-even with daily expenses.

14. [2.5] The slope of the equation is 5, which means the temperature goes up by 5 degrees every hour. So every half hour (30 minutes), the temperature goes up by $0.5 \times 5 = 2.5$ degrees.

15. $\boxed{D}$ Putting the equation into $y = mx + b$ form, $y = \frac{1}{2}x + 7$. The slope of $\frac{1}{2}$ means that one more turtle requires an additional half a gallon of water. So III is true.

    Getting x in terms of y, $x = 2y - 14$. The "slope" of 2 means that 1 more gallon of water can support two more turtles. So I is true.

16. $\boxed{C}$ Because this question is asking for the change in "x" per change in "y" (the reverse of slope), we need to rearrange the equation to get x in terms of C.

$$C = 1.5 + 2.5x$$

    Dividing each element in the equation by 2.5,

$$0.4C = 0.6 + x$$

$$x = 0.4C - 0.6$$

    The slope here is 0.4, which means the weight of a shipment increases by 0.4 pounds per dollar increase in the mailing cost. So a 10 dollar increase in the mailing cost is equivalent to a weight increase of $10 \times 0.4 = 4$ pounds.